# Summary: embeddings and clustering

Data: UC Merced, 2100 aerial images, 21 classes. Train set: 1680 images. Each image has a 768-number embedding from a vision transformer (ViT). Embeddings come from several layers.

## Exercise 1: Distances

- An embedding is a vector of numbers. Close vectors mean similar images.
- Early layers see colour and texture. Deep layers see meaning.
- Cosine similarity compares only the direction of two vectors. It ignores their length.
- Manhattan (L1) distance adds up the absolute differences in each coordinate. Length matters. It assumes all coordinates have a similar scale.
- To compare, take the 10 nearest images of one query image. Check how many share its class. Repeat for layers 1, 6 and 12.

## Exercise 2: k-means (layer 3)

- With 5 clusters, the sizes were 347, 380, 324, 77 and 552.
- Cluster 3 had 77 images. 74 of them (96%) were chaparral. The images look alike: green, with a fine striped texture.
- This cluster is useful. It is almost one class, and k-means never saw the labels.
- The other 4 clusters mix unrelated classes, for example airplanes, beaches, baseball fields and buildings.
- With 10 and 15 clusters and other seeds, the clusters change but the mixing stays. I checked one cluster per run. Each one still mixed buildings, roads, a harbour and houses.
- Reason: layer 3 captures colour and texture, not meaning. Different classes can look similar at this depth.
- A useful cluster is mostly one class. It also stays the same when K or the seed changes.

## Exercise 3: Hierarchical clustering (layer 6)

- The similarity matrix has size 1680 x 1680. Each value is a cosine similarity.
- Most similar pair: images 145 and 147, cosine 1.000. They are almost the same image.
- Most dissimilar pair: images 6 and 578, cosine 0.636.
- I cut each tree into 10 clusters and tried 4 linkage methods.
- **single**: one huge cluster of 1653 images and many tiny ones. This is the chaining effect. Not useful.
- **complete**: balanced sizes. Some pure clusters: agricultural 100% (33 images), chaparral 96% (77), forest 73% (108).
- **average**: some pure clusters (forest 97%, chaparral 100%, harbor 100%, agricultural 97%), but one cluster has 830 mixed images.
- **ward**: the most balanced sizes. Two very pure clusters: chaparral 100% (68 images) and agricultural 99% (69 images). The other 8 are mixed (top class 16% to 43%).
- Meaningful cluster: ward cluster 10. It has 69 images and 99% are agricultural.

## Main points

- Some classes (chaparral, agricultural, forest, harbor) form clean clusters. Many others do not.
- The method matters. Ward gives the best balance. Single linkage fails.
- Layer 3 embeddings gave mostly mixed clusters. Early layers capture colour and texture, not meaning.